

C01-AT3-Coding Challenge

1. Question



SIMATS Online Programming Lab

JAVA PROGRAMMING

5 Challe
Solve Now

CO1-AT3- Coding Challenge

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QUESTIONS

Q1
 Smart University Payroll Management System
(7 marks)

Q2
 Smart College Academic Data System
(3 marks)

(0) answered

Q1 Smart University Payroll Management System
20 points

A university employs three categories of staff: Teaching faculty, Technical staff and Visiting faculty. The payroll system should calculate the monthly salary based on the employee type while maintaining a common employee hierarchy. Requirements Create a Superclass Employee with the following private data members: employment department BasicSalary Provide Data.At constructor Parameterized constructor (salary and tenure methods sharedEmployee() canUseSalary()) Create the following subclasses: Teaching faculty TeachingProfessor InstructorsTechnical Salary BaseSalary < TeachingEmployee < ResearcherInstructor TechnicalStaff researchTechnician everyyearSalary Salary BaseSalary < InstructorInstructor < AcademicInstructor TeachingFaculty TechnicalAndNonAcademicSalary AcademicInstructor < paymentOfTechnicalGrossSalary calculateSalary() in all subclasses. Create an array of Employee objects and demonstrate runtime polymorphism. Display Employee details. Monthly salary (Gross+paid employee Average salary Search an employee Using the employees ID.

```

//Ex.Like 1 (Normal Case)
EmpId
0
TeachingFaculty
101
Dr. Arun
150
Salary
Salary
20
100
TechnicalStaff
102
Mr. Rao
100
10000
15
100
VisitingFaculty
103
Dr. Kumar
10
6
101
Salary
Salary
Research Employee ID:
101
Expected Output:
Employee Details
Employee ID : 101
Name : Dr. Arun
Department : CS
Monthly Salary : 10000.0
Employee ID : 102
Name : Mr. Rao
Department : IT
Monthly Salary : 15000.0
Employee ID : 103
Name : Dr. Kumar
Department : IT
Monthly Salary : 10000.0
Highest Paid Employee
Employee ID : 101
Salary : 10000.0
Average Salary:
43333.33
Employee Found
Employee ID : 102
Name : Mr. Rao
Salary : 15000.0
                
```

1. Answer

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K.Dhrip
TECHNICALSTAFF

Your Code

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int n=Integer.parseInt(sc.nextLine().trim());
6         Employee[] emp=new Employee[n];
7         for(int i=0;i<n;i++){
8             String type=sc.nextLine().trim();
9             while(type.isEmpty()){
10                 type=sc.nextLine().trim();
11             }
12             int id=Integer.parseInt(sc.nextLine().trim());
13             String name=sc.nextLine().trim();
14             String dept=sc.nextLine().trim();
15             double basicSalary=Double.parseDouble(sc.nextLine().trim());
16             int value=Integer.parseInt(sc.nextLine().trim());
17             double value2=Double.parseDouble(sc.nextLine().trim());
18             if(type.equalsIgnoreCase("TeachingFaculty")){
19                 emp[i]=new TeachingFaculty(id,name,dept,basicSalary,value);
20             }
21             else if(type.equalsIgnoreCase("TechnicalStaff")){
22                 emp[i]=new TechnicalStaff(id,name,dept,basicSalary,value);
23             }
24             else if(type.equalsIgnoreCase("VisitingFaculty")){
25                 emp[i]=new VisitingFaculty(id,name,dept,basicSalary,value);
26             }
27         }
28         String temp=sc.nextLine().trim();
29         while(temp.isEmpty()){
30             temp=sc.nextLine().trim();
31         }
32         int searchId;
33         if(temp.equalsIgnoreCase("Search Employee ID:")){
34             searchId=Integer.parseInt(sc.nextLine().trim());
35         }
36         else{
37             searchId=Integer.parseInt(temp);
38             System.out.println("Employee Details");
39             System.out.println();
40             Employee highest=emp[0];
41             double total=0;
42             for(Employee e:emp){
43                 System.out.println("Employee ID : "+e.getEmployeeId());
44                 System.out.println("Name : "+e.getEmployeeName());
45                 System.out.println("Department : "+e.getDepartment());
46                 System.out.println("Monthly Salary : "+e.calculateSalary());
47                 System.out.println();
48                 total+=e.calculateSalary();
49                 if(e.calculateSalary()>highest.calculateSalary()){
50                     highest=e;
51                 }
52             }
53             System.out.println("Highest Paid Employee");
54             System.out.println("Employee ID : "+highest.getEmployeeId());
55             System.out.println("Salary : "+highest.calculateSalary());
56             System.out.println();
57             System.out.println("Average Salary");
58             System.out.println("A.I.F.N",total/n);
59             System.out.println();
60             boolean found=false;
61             for(Employee e: emp){
62                 if(e.getEmployeeId()==searchId){
```

Input

```
1
TeachingFaculty
101
```

Output

```
Employee Details
Employee ID : 101
Name : Dr. Arun
Department : CSE
Monthly Salary : 25000.0
```

2. Question

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K.Dhileep
RESEARCHER

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4/2024

QUESTIONS

Q1
Object-Oriented Program
(Management System)
(5 marks)

Q2
Smart Vehicle Insurance Claim
System
(25 marks)

2/0 answered

Q1 Smart Vehicle Insurance Claim System

25 marks

Problem Statement: An insurance company processes claims for three categories of vehicles: Car, Bike, and Truck. The claim amount depends on the vehicle type. **Requirements:** Superclass: Vehicle Private members: vehicleNumber ownerName insuredAmount Subclasses: Car Bike Truck Each subclass overrides calculateClaimAmount() using different formulas. The application should: Read details for N vehicles. Store them using an array of superclass references. Calculate claim amounts using runtime polymorphism. Search by vehicle number. Display the vehicle with the highest claim amount. **Question 3 (Very Hard):** Online Shopping Order Processing System (50 Marks): **Problem Statement:** An e-commerce company sells products in three categories: Electronics, Clothing, Grocery. The billing process varies for each category. **Requirements:** Superclass: Product Private members: productId productName price quantity Subclasses: Electronics, Clothing, Grocery. Each subclass overrides calculateBill(). The program should: Store products in an array of superclass references. Calculate the final bill using runtime polymorphism. Search products by ID. Display the most expensive order. Compute total revenue.

Claim Formula (Recommended):

Car:

$$\text{Claim} = \text{insuredAmount} \times 0.80$$

Bike:

$$\text{Claim} = \text{insuredAmount} \times 0.60$$

Truck:

$$\text{Claim} = \text{insuredAmount} \times 0.90$$

Test Case 1:

Input:

3

Car:

TN10AB1234
Ramesh
800000

Bike:

TN08XY5678
Suresh
100000

Truck:

TN11TR9999
Logan
1500000

Search Vehicle:

TN08XY5678

Expected Output:

Vehicle : TN10AB1234
Claim Amount : 640000.0

Vehicle : TN08XY5678
Claim Amount : 60000.0

Vehicle : TN11TR9999
Claim Amount : 1350000.0

Highest Claim Vehicle:

TN11TR9999
1350000.0

Vehicle Found:

Owner : Suresh
Claim Amount : 60000.0

2. Answer

